

MAS · Statistical Inference (Sample Mean) — Patterns and Solution Approaches

Basis: 17 past exam questions from 2016–2025, marking key + exam reports. Appears only in Calculator-assumed papers. Original questions: see [topics/07_Statistical_Inference.pdf](#). Numerical examples have been verified with scipy. Also cites 2016 sample paper A9, A13 (original papers: see [papers/2016Sample_*](#)). Review status: has undergone one round of independent review and has been revised according to feedback (see [Review_log.md](#)).

Formula card

- Sample mean: $\bar{X} \sim N(\mu, \sigma^2/n)$ (approximate). If the population itself is normal, it is exact for any n ; if the population is not normal, by the Central Limit Theorem it is approximately true when n is large enough (empirically ≥ 30), regardless of the population shape.
- Standard deviation of the sample mean (standard error): $\sigma_{\bar{X}} = \sigma/\sqrt{n}$.
- Sum of n variables: key's method is to convert "sum < T" into " $\bar{X} < T/n$ " (2016A-Q19(b), 2020A-Q18(a)). $\text{Sum} \sim N(n\mu, n\sigma^2)$ is only used for checking; key does not give this form.
- Confidence interval for μ : $\bar{x} \pm z \cdot \sigma/\sqrt{n}$ (when σ is unknown, use sample standard deviation s instead); $z = 1.645 / 1.960 / 2.576$.
- Working backwards from an interval: $\bar{x}$ = interval midpoint; $z \cdot \sigma/\sqrt{n}$ = half-width $\Rightarrow$ can solve for $\sigma/\sqrt{n}$, s or n .

Marking panel reminders

- **Write mathematical statements, not calculator language** (2016, 2017, 2018 reports): write $P(X > 83) = P(Z > 1.5) = 0.0668$, not `normCdf(83, ∞, 80, 2)`.
- **"State the distribution" 3 marks**: ✓ state that X is approximately normal (writing $n > 30$ or CLT as well is safer; in each year's key the first mark only requires writing "normal") ✓ mean ✓ standard deviation $\sigma/\sqrt{n}$ (or variance).
- **When explaining, make clear which quantity is meant**: "the standard deviation of the sample mean decreases", not just "standard deviation decreases", and do not use "it" (2019 report).
- **Answer the question asked**; do not copy sentences from previous years' standard answers (2025 report).

Pattern 1: Write the distribution of $\bar{X}$ and find probabilities ★ High frequency

Standard procedure

1. "By the Central Limit Theorem, since $n = 100$ is large, $\bar{X}$ is approximately normal with mean $\mu = 80$ and standard deviation $\sigma/\sqrt{n} = 20/10 = 2$."
2. Write the probability expression, then calculate.
3. "differs from μ by less than d " = $P(|\bar{X} - \mu| < d) = P(\mu - d < \bar{X} < \mu + d)$.
4. Sum type: 50 times total < 8.96 kL $\Leftrightarrow \bar{X} < 179.2$ L $\Rightarrow \bar{X} \sim N(175, 2.1213^2) \Rightarrow P \approx \mathbf{0.9761}$ (2016A-Q19(b)).

When the population is not normal (uniform distribution 2017A-Q9, exponential distribution 2018A-Q12, right-skewed pdf 2021A-Q15, dart distance 2025A-Q13): the method is the same as for a normal population, because n is large enough, $\bar{X}$ is still approximately normal. 2018A-Q12(c) specifically tests this (2 marks: answer unchanged + reason is CLT). **Drawing the distribution of $\bar{X}$** : a bell shape centred at μ , width about $\pm 3\sigma/\sqrt{n}$, much narrower than the original distribution; if the question requires it, mark on the diagram the probability region calculated in the previous part (2017A-Q9(b) has 1 mark, 2021A-Q15(b)).

Past exam questions: 2016A-Q19(a)(b), 2017A-Q9, 2018A-Q12(a)(b), 2019A-Q14(a)(b), 2020A-Q18(a), 2021A-Q15(a), 2022A-Q12(a)(b), 2023A-Q13(a)(b), 2024A-Q14(a)(b), 2025A-Q13(a)(b).

Pattern 2: Find the sample size n (so that a probability or interval meets a requirement)

Standard procedure

1. First draw a diagram to decide whether it is two-tailed, one-tailed or half-interval, then write the probability statement. Two-tailed: $P(|X - \mu| < 0.2) = 0.98$; one-tailed: $P(X > 25) = 0.03 \Rightarrow z = 1.881$ (2018A-Q12(d)); half-interval: $P(80 < X < 82) > 0.4 \Rightarrow P(0 < Z < k) = 0.4$, $k = 1.282$, $n = 165$ (2019A-Q14(d)).
2. Find the critical z ($98\% \Rightarrow z = 2.326$).
3. Set up an equation or inequality: $z \cdot \sigma / \sqrt{n} = d \Rightarrow n = (z\sigma/d)^2$.
4. **Rounding**: when "at least" is required, round up. Example: $(2.326 \times 1.5/0.2)^2 = 304.3 \Rightarrow n = 305$ (2020A-Q18(b)); $600.25 \Rightarrow 601$ (2024A-Q14(f), when "at least" is required, round up no matter how small the decimal part is). When working backwards from a known interval to find n , take the nearest integer (2022A-Q14(a): using $z = 2.576$ gives $200.0 \Rightarrow 200$, key also accepts using 2.58 to get 201 ; 2025A-Q13(c): $283.0 \Rightarrow 283$, also accepts 284).

Past exam questions: 2016A-Q19(d), 2018A-Q12(d), 2019A-Q14(d), 2020A-Q18(b), 2022A-Q14(a), 2024A-Q14(f), 2025A-Q13(c); 2016S-A9, 2016S-A13(b) (sample paper). Finding the error bound in reverse: 2024A-Q14(c) (50% probability corresponds to $m \approx 1.7$ minutes).

Pattern 3: Confidence interval calculations and working backwards

Calculation: $\bar{x} \pm z \cdot s / \sqrt{n}$. Example: $\bar{x} = 10300$, $s = 400$, $n = 100$, $95\% \Rightarrow 10300 \pm 1.96 \times 40 = (10221.6, 10378.4)$ (2017A-Q13(a): 1 mark each for centre, standard error, z , upper and lower limits). **Working backwards**

- Upper limit 40.62, width 1.08 $\Rightarrow \bar{x} = 40.08$; $0.54 = 2.576 \times s/20 \Rightarrow s \approx 4.19$ (2018A-Q16).
- Interval $150 \leq \mu \leq 200 \Rightarrow \bar{x} = 175$, $\sigma_{\bar{X}} = 25/1.96 \approx 12.76$; after doubling the sample size, $\sigma_{\bar{X}} = 12.76/\sqrt{2} \approx 9.02 \Rightarrow P(|X - \mu| < 10) \approx 0.7325$ (2020A-Q17).
- Given the interval and n , find the confidence level in reverse: first find $z = \text{half-width} / \text{standard error}$, then look up the corresponding percentage (2017A-Q13(d)).
- Changes in sample size: width $\propto 1/\sqrt{n} \Rightarrow$ to make the width $1/4$, $16n$ is needed (2021A-Q17(c)); if the sample size becomes $2n$, the standard error is divided by $\sqrt{2}$.

Past exam questions: 2016S-A13, 2017A-Q13, 2018A-Q16, 2019A-Q15, 2020A-Q17, 2021A-Q17, 2022A-Q12(c), 2023A-Q13(c)(d), 2023A-Q15, 2024A-Q14(d).

Pattern 4: Interpreting confidence intervals (true/false) ★ tested almost every year (except 2016, 2020)

Correct statements to memorise

1. μ is a **fixed but unknown** number. A particular interval either contains μ or does not, and **there is no way to know which** (2018A-Q16(d), 2019A-Q15(d), 2021A-Q17(e), 2022A-Q14(c)).
2. "95% confidence" means that when sampling is repeated, about 95% of intervals will contain μ ; **it is not** "the probability that μ lies in this interval is 95%" (2023A-Q15(c): Ali's statement is incorrect).
3. A wider interval does not mean it is more likely to contain μ (2022A-Q14(c): James is incorrect).
4. The mean of the next sample **does not necessarily** fall in this interval (2017A-Q13(b)(i)); an individual value is even less likely to, because the variation of individuals is much greater than the variation of sample means (2017A-Q13(b)(ii)).
5. Having 10 out of 50 intervals with a lower limit below 1.00 does not indicate that the setup is wrong; this is normal variation from random sampling (2023A-Q13(e)). **Conversely**: if, among a large number of intervals, the proportion that do not contain μ is far above 5%, that is evidence. In 2025A-Q13(e), 298 of 500 95% intervals do not contain $\mu = 100$; key judges Chen's statement 1 (the sample is unlikely to come from a population with $\mu = 100$) **correct**; but the cause still cannot be determined, so

statement 2 (the athlete did not stand at 3.50 m) is **incorrect**.

6. Two intervals being nested or overlapping **does not** imply that the two population means are the same; nor can a lower sample mean alone be used to say the new method is better (2024A-Q14(e); key's reason for Sanjeet is: the two intervals are based on different sample sizes, so they cannot be compared directly in this way).
7. Duplicating **the same sample** to make $2n$ is not a new random sample, does not reflect real variation, and the interval does not truly become narrower (2022A-Q12(d)). Note: combining two **independent** random samples is valid; n is larger and precision is higher (2019A-Q15(c)).

Writing style: first state a position (True / False / Not correct), then give one reason in the context of the question; 1 mark each.

Pattern 5: Using an interval to test a claim (does a company use more water?)

Standard procedure (4 marks)

1. From the new sample, find $\bar{x}$ and s : $6.57 \text{ kL} / 36 = 182.5 \text{ L}$, $s = 15$.
2. Construct a confidence interval for μ_{new} : $182.5 \pm 1.96 \times 15/6 = (177.6, 187.4)$.
3. Compare: the original $\mu = 175$ **is not** in the interval.
4. Conclusion: there is evidence to support "WolliWorks uses significantly more water" (2016A-Q19(e)); conversely, if it is in the interval, there is not enough evidence (2020A-Q18(c)).

Note: you can only draw a conclusion about "whether the means are different"; you cannot infer the source of the sample (for example, Anika's claim "probably taken from 100 teenagers"), because the source is unknown (2021A-Q15(c)).

Past exam questions: 2016A-Q19(e), 2020A-Q18(c), 2021A-Q15(c), 2025A-Q13(d)(e).

Pattern 6: Factors affecting precision

Change	Standard deviation of $\bar{X}$	Interval width	Tail probabilities such as " $X > 83$ " (2019A-Q14: $\mu = 80$, threshold 83 is above μ)
n increases	decreases	becomes narrower	decreases (83 is relatively more extreme)
Confidence level increases	—	becomes wider	—
σ (or s) increases	increases	becomes wider	increases

The larger the sample and the smaller the standard error, the higher the precision (2019A-Q15(c): the combined sample $3n$ has the highest precision). To compare which of two intervals is narrower: compare $z \cdot s / \sqrt{n}$ (2021A-Q17(f)).